

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 2: GAME-BASED LEARNING (5 QUESTIONS)

Course Code:	MLA0407	Course Title:	Deep Learning
CO Assessed:	CO3	Assessment:	Assessment 2: Game-Based Learning
Weightage:	35% of CIA	Total Marks:	(5 Questions × 10 Marks)
CO1 Statement:	<i>Apply the fundamental concepts of machine learning and neural networks to design and analyze learning algorithms using perceptrons, multilayer perceptrons, forward propagation, backpropagation, gradient descent optimization techniques, and Maximum Likelihood Estimation for solving real-world problems.</i>		
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Section / Batch:	AIML -2024	Date:	20.08.2026

Code of Conduct

I CH. Durga Prasad (192425305) certify that this submission is my original work and that I have adhered to the guidelines specified this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Ch. Durga Prasad

QUESTIONS: 5

Total Marks: 50

Annexure A

Game-Based Learning

Game Title: CNN Quest – The Image Intelligence Challenge Game

Scenario:

Students act as AI Engineers responsible for building a CNN-based image classification system. They must complete different levels by correctly applying CNN concepts. Each level tests a different part of the CO3 syllabus.

Level 1 – CNN Fundamentals

1. What is the primary purpose of a CNN in image processing, and how is it different from a traditional Artificial Neural Network (ANN)?
2. Arrange the following CNN components in the correct order:
Convolution Layer → Pooling Layer → Fully Connected Layer → Input Image → Output/Classification.
3. Identify the CNN component:
"I scan across an image and detect patterns such as edges and textures. I use the same set of weights at different locations." Who am I?

Level 2 – Convolution & Filters

4. A 5×5 image is processed using a 3×3 kernel with a stride of 1 and no padding. What will be the size of the resulting feature map? Explain how you obtained the answer.
5. A CNN uses different kernels to detect edges, curves, textures, and other patterns. Explain why a CNN requires multiple filters instead of using only one filter throughout the network.
6. Match the following:

Component	Function
Kernel	?
Feature Map	?
Stride	?
Padding	?

7. Students must correctly match each component with its function.

Level 3 – Feature Map & Pooling Challenge

7. A CNN produces a feature map containing strong activation values around the edges of an object.
What does this feature map tell us about the image, and why are feature maps important in CNNs?
8. A 4×4 feature map is given below:
9. 1 3 2 4 10. 5 6 1 2
11. 7 2 8 3
4 5 6 9

Apply 2×2 Max Pooling with a stride of 2 and calculate the resulting feature map.

12. Max Pooling vs Average Pooling:
Students are given different scenarios and must choose which pooling method would be more appropriate and justify their choice.

Level 4 – CNN Architecture Challenge

10. A CNN model performs extremely well on training images but gives poor results on new images.

Identify the problem and select the most appropriate solution from the following:

- Increase overfitting

- Remove all training data
- Use regularization/data augmentation
- Increase the image noise Explain your choice.

11. Parameter Sharing Puzzle:

A convolution filter contains 9 weights and is applied across different regions of an image.

Explain how the same 9 weights can be reused at different locations and why this significantly reduces the number of parameters in a CNN.

Level 5 – Architecture Battle: AlexNet vs ResNet

12. Students are shown two simplified CNN architectures and must identify which represents AlexNet and which represents ResNet.

13. AlexNet vs ResNet Challenge:

Match the characteristics with the correct architecture:

- Introduced residual/skip connections
- Famous for ImageNet performance
- Helps solve vanishing-gradient problems in deep networks □ Uses a relatively deep CNN architecture for its time

14. Scenario Challenge:

You are given a complex image-classification problem requiring a very deep CNN.

Would you choose AlexNet or ResNet? Why?

Students must provide a justification based on architecture and training considerations.

Level 6 – CNN Real-World Mission

15. Students are given different applications:

- Face Recognition
- Self-Driving Cars
- Medical Image Diagnosis
- Handwritten Digit Recognition
- Object Detection
- Agricultural Disease Detection

Match each application with the image-processing task performed by CNNs.

16. Final Boss Challenge :

Design a basic CNN architecture for a cat-vs-dog image classification system.

Students must specify:

- Input image
- Number/type of convolution layers
- Filters
- Pooling layers
- Fully connected layer
- Output layer
- Classification result

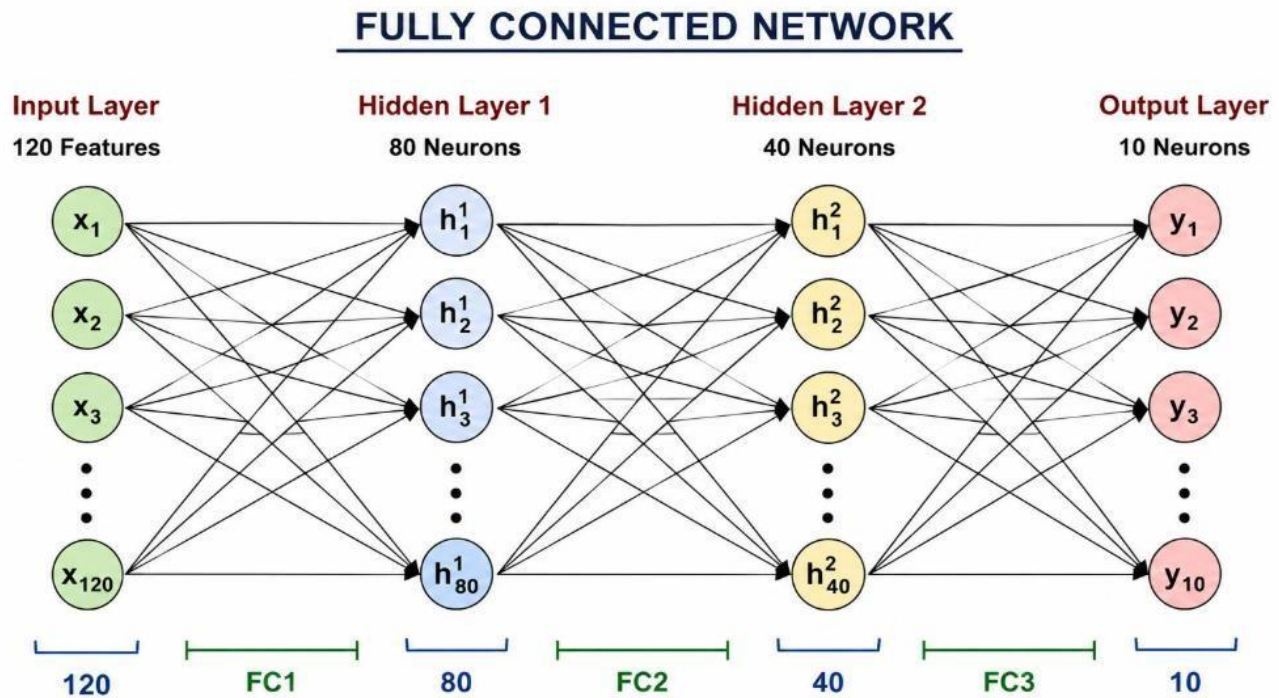
COMPLEX QUESTION 1 – FULLY CONNECTED LAYER Problem

A fully connected neural network receives an input vector of **120 features** and passes it through three dense layers:

- FC1: $120 \rightarrow 80$ neurons

- FC2: $80 \rightarrow 40$ neurons
- FC3: $40 \rightarrow 10$ neurons Each neuron has one bias.

1. Calculate the Total Number of Trainable Parameters A fully connected layer connects every neuron in one layer to every neuron in the next layer. **Network Structure**



Every input is connected to every neuron

Formula

Number of parameters:

Parameters = Weights + Biases

Weights = Number of inputs $\times$ Number of neurons

Biases = Number of neurons

FC1: $120 \rightarrow 80$

Weights: $120 \times$

$80 = 9,600$

Biases:

80

Therefore:

FC1 Parameters = $9,600 + 80 =$

9,680

FC2: $80 \rightarrow 40$

Weights: $80 \times$

$$40 = 3,200$$

Biases:

$$40$$

Therefore:

$$\text{FC2 Parameters} = 3,200 + 40 = 3,240$$

FC3: $40 \rightarrow 10$

Weights: $40 \times$

$$10 = 400$$

Biases:

$$10$$

Therefore:

$$\text{FC3 Parameters} = 400 + 10 = 410$$

Total Parameters

$$\text{Total} = 9,680 + 3,240 + 410 = 13,330$$

Therefore:

TOTAL TRAINABLE PARAMETERS = 13,330

2. Number of Multiplications and Additions

During a forward pass, each connection performs a multiplication.

FC1

$$120 \times 80 = 9,600 \text{ multiplications}$$

FC2

$$80 \times 40 = 3,200 \text{ multiplications}$$

FC3

$$40 \times 10 = 400 \text{ multiplications}$$

Total multiplications:

$$9,600 + 3,200 + 400$$

$$= \mathbf{13,200 \text{ multiplications}}$$

Additions

The weighted inputs must be added together.

Ignoring bias:

- FC1 = 9,600 additions
- FC2 = 3,200 additions

- FC3 = 400 additions

Total = 13,200 additions

There are also 130 bias additions:

$$80 + 40 + 10 = 130 \text{ Therefore:}$$

Total additions including bias:

$$13,200 + 130$$

= 13,330 additions Summary Layer Multiplications Additions

FC1	9,600	9,680
FC2	3,200	3,240
FC3	400	410
TOTAL	13,200	13,330

3. FC2 Changed from 40 to 60 Neurons

The new architecture becomes: 120

Inputs



80 Neurons



60 Neurons



10 Neurons

Changing FC2 affects **both FC2 and FC3**, because FC3 now receives 60 inputs instead of 40.

FC1: 120 → 80 Parameters:

$$(120 \times 80) + 80 = 9,680$$

FC2: 80 → 60 Parameters:

$$(80 \times 60) + 60 = 4,800 + 60 = 4,860$$

FC3: 60 → 10 Parameters:

$$(60 \times 10) + 10 = 600 + 10 = 610$$

New Total

$$9,680 + 4,860 + 610$$

= 15,150 Therefore:

NEW TOTAL TRAINABLE PARAMETERS = 15,150

Increase: 15,150

– 13,330

= 1,820 additional parameters

4. Effect of Increasing FC2 Neurons

Increasing the neurons from 40 to 60 increases the capacity of the neural network. FC2:

40 Neurons



Smaller Model



Less computation



Lower memory usage

FC2: 60 Neurons



Larger Model



More parameters



More computation



Higher memory usage



Can learn more complex patterns



Higher risk of overfitting

Advantages

- Can learn more complex relationships.
- Provides greater model capacity.
- May improve accuracy if the original network was underfitting.

Disadvantages

- More parameters.
- More computation.
- More memory required.
- Training can take longer.
- Increased risk of overfitting.

Conclusion

Changing FC2 from **40 to 60 neurons** increases the parameters from **13,330 to 15,150**. The larger network can learn more complex patterns, but regularization techniques such as **Dropout and L2 regularization** may be required to control overfitting.

COMPLEX QUESTION 2 – PARAMETER SHARING Problem

A CNN processes an RGB image of size:

$64 \times 64 \times 3$ It

uses:

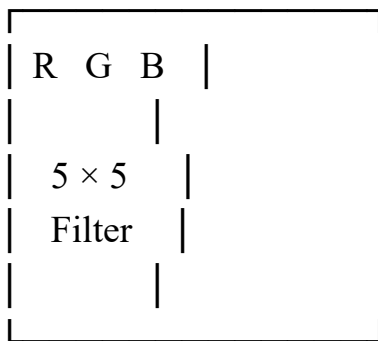
- Number of filters = 32
- Filter size = 5×5
- Stride = 1
- Padding = 0

1. Calculate Trainable Parameters Using Parameter Sharing Because the input is an **RGB image**, it has 3 channels.

Therefore, each filter must have depth 3.

One Filter

RGB Image



Filter Size = $5 \times 5 \times 3$ Number of weights:

$$5 \times 5 \times 3 = 75$$

Each filter also has one bias.

Therefore:

$75 + 1 = 76$ parameters per filter For 32 filters:

76×32

= 2,432 Therefore:

TOTAL TRAINABLE PARAMETERS = 2,432

2. Calculate Output Feature Map Dimensions Formula:

Output Size = ((Input Size – Filter Size) / Stride) + 1 Substitute the values:

Output Size = $((64 - 5) / 1) + 1$

$= 59 + 1$

= 60

So each filter produces a:

60×60 feature map

Since there are 32 filters: **Output**

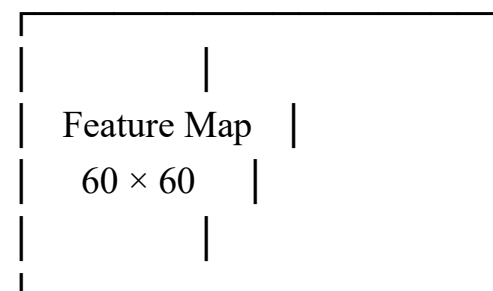
$= 60 \times 60 \times 32$

Visual Explanation

Input Image

$64 \times 64 \times 3$

|
| 5 × 5 Filter



|
|
32 Filters

▼
 $60 \times 60 \times 32$
Output

3. Total Number of Neuron Activations

Output dimensions: $60 \times 60 \times 32$

Therefore:

$$60 \times 60 = 3,600$$

$$3,600 \times 32 = \mathbf{115,200}$$

Therefore:

$$\mathbf{\text{TOTAL NEURON ACTIVATIONS} = 115,200}$$

4. Parameters Without Parameter Sharing

In a CNN, the same filter is normally reused at different spatial locations.

This is called **parameter sharing**.

Without parameter sharing, every spatial location would have its own separate set of weights.

Number of Spatial Locations

$$60 \times 60$$

$$= 3,600 \text{ locations}$$

Each location requires: 5

$$\times 5 \times 3 = 75 \text{ weights}$$

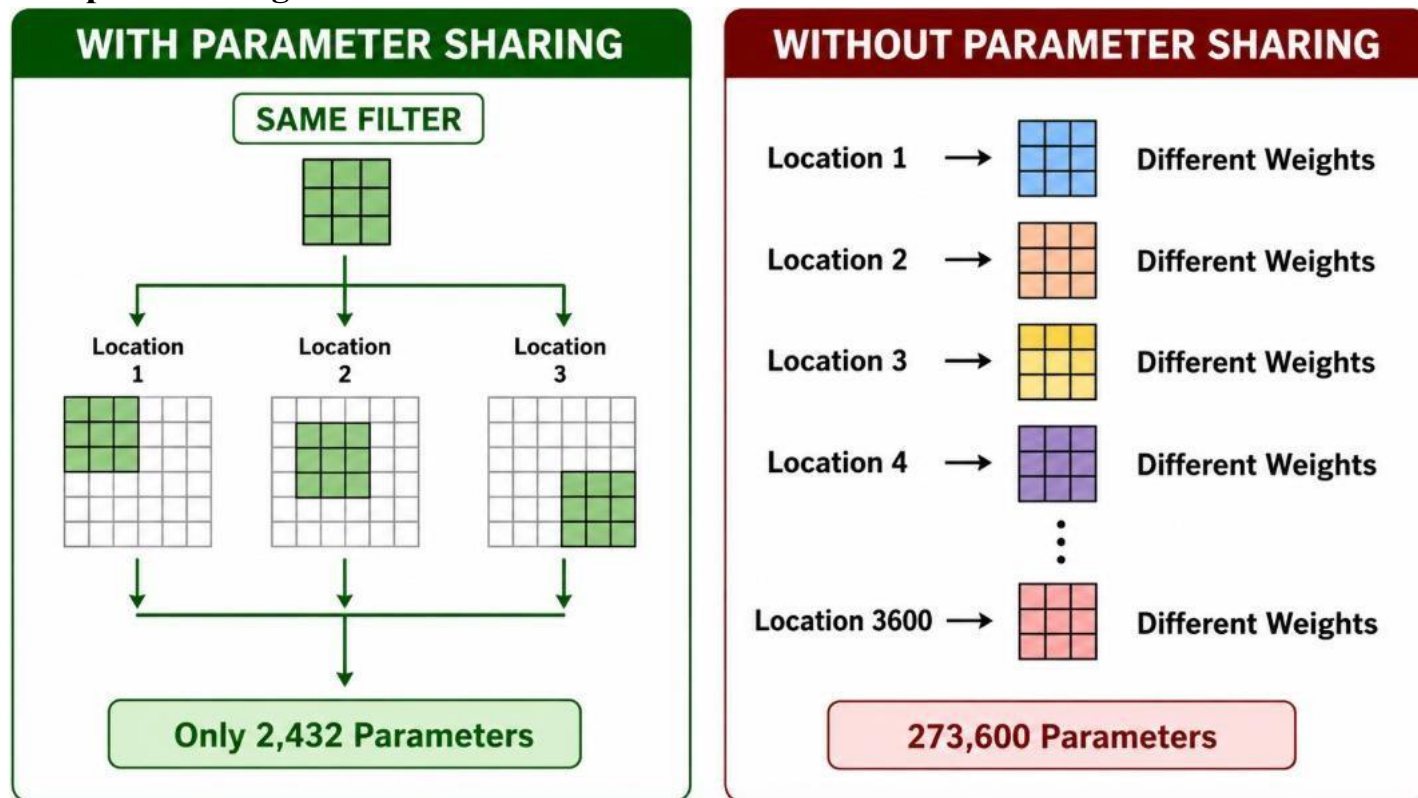
Plus 1 bias:

$$75 + 1 = 76 \text{ parameters}$$

$$\text{Therefore: } 3,600 \times 76$$

= 273,600 parameters

Comparison Diagram



5. Comparison of Both Approaches

Feature	With Parameter Sharing	Without Parameter Sharing
Parameters	2,432	273,600
Memory	Low	Very High
Computation	Efficient	More Expensive
Training	Faster	Slower
Overfitting	Lower	Higher
Generalization	Better	Lower
Feature Detection	Same feature anywhere	Location-specific

Memory Usage

Parameter sharing requires only **2,432 trainable parameters**, while no parameter sharing requires **273,600 parameters**.

Therefore, parameter sharing greatly reduces memory requirements.

Computational Efficiency

With parameter sharing, the same filter is reused throughout the image. One Filter



Location 1

Location 2

Location 3

Location 4

...

Location 3600

This makes CNNs much more efficient than assigning separate parameters to every location.

Generalization

Parameter sharing allows the network to recognize the same feature regardless of where it appears in the image. For example:

Edge on left side → Detected

Edge in center → Detected Edge

on right side → Detected

The same filter can detect the edge in all these locations.

This improves **generalization** and reduces the risk of overfitting.

FINAL CONCLUSION

Fully Connected Network

The original network contains:

13,330 trainable parameters

After changing FC2 from 40 to 60 neurons:

15,150 trainable parameters

The increase improves model capacity but also increases computation and overfitting risk.

CNN Parameter Sharing With

parameter sharing:

2,432 parameters

Without parameter sharing:

273,600 parameters

Thus, parameter sharing is one of the major reasons CNNs are efficient. It significantly reduces the number of parameters, memory usage, and overfitting while allowing the same feature detector to work across different locations of an image.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, wellanalyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: _____ Signature: _____ Signature: _____ Date: _____ Date: _____ Date: _____		